

Personal Knowledge Base — RAG Profile

A structured personal document for use as a RAG / AI knowledge source.

1. Basic Information

Full Name: Abdullah Firoj

Preferred Name: Firoz

Date of Birth: 22 june

Current Location: Dhaka, Bangladesh

Nationality: Bangladeshi

Email: abdullahfiroj860@gmail.com

Phone:

Portfolio:

GitHub: [Afiroj313501](#)

LinkedIn: [abdullah-firoj-900697375](#)

2. Education

University: United International University (UIU)

Degree: B.Sc. in Computer Science and Engineering

Study Period: March 2022 – July 2026

College: Ghatail Cantonment College, Tangail

HSC: Science, 2017 – 2019

SSC Institution: BBK HIGH SCHOOL

SSC Group: SCIENCE

SSC Year: 2017

3. Career & Professional Profile

Current professional title: Full Stack Engineer / Software Developer.

Main interests: Software Development, Full-Stack/MERN/PERN, AI/ML, RAG, Generative AI, Data, MLOps, Cybersecurity, and Research.

Goal: Start a software/technology career as soon as possible and grow into a strong technical professional.

Experience: Freelance Software Developer, Self-Employed, 2025 – Present.

Research interests include Agentic AI, Model Context Protocol (MCP), RAG, AI-assisted systems, and multimodal deepfake detection.

Preferred work/industry:

Dream company: OWN COMPANY

Long-term career goal: ITS SECRET

4. Technical Skills

Programming: Python, Java, SQL

Frontend: React, JavaScript, HTML, CSS

Backend: Node.js, Express.js

Database: PostgreSQL, pgvector

ORM: Prisma

AI/ML: Generative AI, RAG, Gemini API, TensorFlow, PyTorch, OpenCV, Scikit-learn, Pandas, NumPy

Other: REST APIs, JWT authentication, Git/GitHub, Vercel, Render, Supabase

Other skills to add:

5. Major Projects

ResearchPilot AI

AI-powered research assistant using PERN + RAG, semantic document retrieval, embeddings, pgvector, Gemini, and citation-grounded Q&A.

Live Demo: <https://research-pilot-orcin.vercel.app>

Repository: <https://github.com/Afiroj313501/ResearchPilot>

LumenLearner / LumosLearn

PERN-based learning management platform with Admin/Instructor/Student roles, JWT authentication, lessons, enrollment, progress, quizzes, assignments, grades, Gemini and RAG.

Live Demo: <https://lumenlearner-iota.vercel.app>

Repository: <https://github.com/Afiroj313501/LumosLearn>

Onboarding Platform

Web-based onboarding platform.

Live Demo: <https://onboarding-platform-xi.vercel.app>

Repository: <https://github.com/Afiroj313501/onboarding-platform>

IMH

Extensive Medical Hub project.

Live Demo:

Repository: <https://github.com/Afiroj313501/IMH>

Other projects:

Favorite / most meaningful project and why:

6. Research & Academic Work

Research paper: "Energy-Aware Tool Discovery for Agentic AI with the Model Context Protocol"

Target venue/style: IEEE Transactions on Sustainable Computing.

FYDP: Multimodal Deepfake Detection (Audio and Video Analysis) — Human in the Loop.

FYDP dataset expansion and model results: accuracy 86.0%, ROC-AUC 0.88, Fake Recall 84.0%, Real Recall 88.1%.

Other research topics:

Research publications:

7. Hobbies & Personal Interests

Hobbies: Football , Video Games , Cooking , Gardening , Listening Music

Favorite activities:

Favorite sport: Football

Favorite football player: Lionel Messi

Favorite football club: FC Barcelona

Favorite national team: Argentina

Favorite football memory: LEO LIFT FIFA WC TROPHY IN LUSAIL

Favorite non-football sport: F1

Favorite games/video games: eFootball

Favorite books: Alchemist , 48 laws of power, Paradoxical Sajid

Favorite movies: Forrest gump, fight club , john wick, shawshank redemption,

Favorite TV series: Breaking bad , Narcos, GOT,

Favorite anime: didnt see any anime

Favorite superhero: BATMAN , IRONMAN

Favorite superhero universe: MCU

Favorite fictional character: BATMAN

Favorite actor: KEANU REEVES

Favorite actress: SADIE SINK (Currently)

Favorite director: C.Nolan

Favorite musician/band: Pink Floyd / Linkin Park / Nirvana

Favorite singer: Chester , Abel (from weekend), Bob dylan

Favorite songs: High hopes, ONE MORE LIGHT , Bhool , Hum, November rain,

Favorite album: Hybrid Theory

Favorite genre of music:

Favorite perfume/fragrance: LV IMAGINATION, CREED AVENTUS ABSULO, JPG LA MALE,

Favorite food: Biryani

Favorite drink: Water

Favorite color: WHITE

Favorite place : MY HOME

Dream travel destination: SWITZERLAND

10. Favorite Things — Fill In

Favorite color: WHITE

Favorite number: 13

Favorite animal: PANDA , BABY COW

Favorite car: BMW

Favorite motorcycle: ROYAL ENFIELD CLASSIC

Favorite quote: VENI VEDI VICI